

Analysis of Trans-membrane proteins using DFT Techniques with a view to predicting function and structure

ABSTRACT

The biological activity of a protein does not manifest itself until the protein folds into its 3-dimensional structure. One of the most fascinating demonstrations of nature's laws is how protein structure is fundamentally directed by the underlying amino acid sequence and the interplay in the residues' chemical interactions. In this project, I have explored the question of whether spectral data can guide the structural and functional analyses of Trans-membrane (TM) proteins. I have produced some initial results using DFT techniques and hydrophobicity scales that demonstrate that TM regions comprising alpha-helices have characteristic frequency components very distinct from those seen in beta-dominated TM regions. In addition, I have confirmed that the characteristic frequencies of the alpha and beta strands are around .29 and .46 respectively correlating quite well with the known structural periodicities of 3.6 and 2.3 residues per turn respectively observed in nature. An attempt has also been made to better understand the relationship between the frequency spectrum and the structural and functional properties of two specific proteins – 1c17 and 2a9h.

Finally, new avenues for research have been suggested. These include performing an exhaustive analysis of all known TM protein structures; detailed analysis of the spectrum of a particular TM protein to better understand which frequency components are due to noise and which embed real insights; new research in the choice of numerical scales for the DFT mapping; and finally incorporating DFT techniques along with probabilistic and hydropathy-based analysis into a comprehensive structure prediction framework.

INTRODUCTION

The biological activity of a protein does not manifest itself until the protein folds into its 3-dimensional structure. One of the most fascinating demonstrations of nature's laws is how protein structure is fundamentally directed by the underlying amino acid sequence and the interplay in their chemical interactions. Neighboring amino acids interact to give rise to first order ("secondary") structures called α -helices followed by more distant interactions leading to β -sheets. Additional interactions lead to the tertiary 3-D folds. As a direct result of the above phenomena sequence analysis is one of the most useful and arguably one of the most exciting methods for protein structure prediction.

Overview of Protein Structures

Proteins are chains of amino acids, with the amino acids characterized by a unique side chain called the R group. The various components of each individual amino acid play a role in the overall structure of the protein. The ability of the chain to rotate around the C α atom; the polarity of the amino acid with the bonding ability of the C=O group (acting as an H bond acceptor) and the NH group (H bond donor); and the particular chemistry of the R-group all drive the formation of secondary structures, folds, and overall 3-dimensional structure. At a fundamental level the protein core comprises regularity in the form of α -helices, β -sheets, loops and coils. α -helices tend to be found on the surface and are readily identified by structures with a periodicity of 3.6 amino acids per turn with an H-bond between every 4th residue. β -sheets on the other hand are formed by chains of 5-10 residues connected by H-bonds with the chains running in parallel or anti-parallel directions. Hairpin loops tend to be found between α -helices and β -sheets typically on the surface of structures. Regions that cannot be classified as α -helices, β -sheets or loops are denoted as coils. Recent analyses of sequences have shown the existence of motifs and domains where a motif indicates the presence of an individual folded structure lying in the core of the protein. Modules on the other hand are a more fundamental unit related to specific units of function. An example of a motif is the C₂H₂ zinc finger DNA-binding motif Xfin of *Xenopus laevis*[4].

TM proteins are a uniquely interesting class of proteins both for their structural and functional properties. By spanning the lipid bilayer they are able to mediate nearly all functional aspects of the membrane including signaling, transport, and adhesion. Structurally, most TM proteins traverse the lipid bilayer as a series of alpha-helical conformations. Beta-barrel multi-pass TM proteins are also known to exist in the outer membrane of mitochondria, bacteria, and chloroplasts [25]. A key property of TM proteins is their amphipathic character with their hydrophobic regions residing in the interior of the bilayer and the hydrophilic regions exposed to water on either side.

Need for Structure Prediction Methods

As of September 2007 there were 42,073 proteins in the Protein Data Bank (PDB) whereas the UniProt/SwissProt release contained 263,525 sequences. A ratio of one structure for every 6 sequences! Even after many decades of research, overall accuracy of

structure prediction is only at around 65-70% [4]. There are many reasons cited for this relatively low level of accuracy including the presence of non-local residue interaction [8] and environment-specific effects [24] that are not well modeled in prediction algorithms due to their inherent algorithmic complexity. However, the possibility that these effects can be elegantly captured in the frequency domain has inspired research in Discrete Fourier Transform (DFT) techniques. Research has shown that distinct frequencies in hydrophobic content of amino acid sequences appear to be well correlated with certain secondary structural elements [1][2].

The high degree of structural periodicity in TM proteins with alpha-helical regions of length 19-20 residues lend themselves well to DFT analysis techniques. Hydrophobicity analysis has already been used quite successfully in their analysis [26]. The motivation for this paper and the rationale for DFT techniques were born out of the need to find efficient techniques to capture non-local residue interactions and environment effects that were not completely captured using local window-based and occurrence-frequency based techniques. In addition it is believed that DFT could more elegantly capture the structural periodicity that is so manifest in these natural structures.

RESULTS

DFT Techniques

This project used the following DFT formulation [23]

$$x[i] = \sum ReX' [k] \cos(2\pi i/N) + \sum ImX' [k] \sin(2\pi i/N)$$

where ReX' and ImX' are the *sinusoidal* amplitudes and are related to the Frequency domain of the signal by the following relations

$$ReX'[k] = ReX[k] / (N/2)$$
$$ImX'[k] = - ImX[k] / (N/2)$$

Except for the special cases:

$$ReX'[0] = ReX[0] / N$$
$$ReX'[N/2] = ReX[N/2]/N$$

The DFT is computed using

$$Re X[k] = \sum_i x[i] \cos(2\pi ki/N) \quad i = 0..N$$
$$Im X[k] = -\sum_i x[i] \sin(2\pi ki/N)$$

$$Amplitude = (Re X[k]^2 + Im X[k]^2)^{1/2}$$

In the above, N represents the length of the amino acid sequence, and $x[i]$ are the hydrophobicity values of the amino acids (Table 2). The choice of hydrophobicity values is described in detail in the next section.

In computing and coding the DFT several implementation-specific issues must be considered especially related to the boundary conditions (see above). These special cases are nicely laid out in [23] and greatly aided in my implementing a correct DFT for the purposes of this project. It is unfortunate that academic papers invariably use the complex form of the Fourier Transform which isn't readily amenable to software implementation. Given that in this term project, the DFT implementation was a critical part of the effort, I felt it was important to highlight some of these issues.

Rationale behind choice of Hydrophobicity Values

The choice of hydrophobicity scales is of critical importance in detecting periodicity and is the subject of much debate. The "Prior Work" section provides a flavor of some of the ongoing research in this area as well as the controversy regarding the usefulness of hydropathy for structure prediction.

Numerous scales have been developed to measure the hydrophobicity/hydrophilicity of amino acids. Some scales [6] are derived from examining proteins with known 3-D structure and assigning values depending on an amino acid's propensity to be in the core. Other scales such as the Kyte&Doolittle scale [3], are derived from the physico-chemical properties (water solubility, effect on surface tension etc) of the residues. A rigorous analysis of 38 of such scales was done in [1] using the concept of an amphipathic index to evaluate the predictive quality of the various scales.

Based on those evaluation results as well as my own experiments during this project, I standardized on the Kyte&Doolittle hydrophobicity scale for my experiments (see Table 2). The use of alternate scales, such as Hopp-Wood, EIIP-based, Eisenberg, Engelman's GES, or other custom numeric schemes did not produce significantly superior results.

Zero Frequency Adjustment

Early on in my experiments I observed a significantly high zero frequency component in the DFT that was drowning out other important periodicities. The reasons for this phenomena can be several from the possible long-range power-law or $1/f^\beta$ behavior correlations phenomena observed in [16] to the issues related to the finite length of the sequence under consideration as shown in Figure 8. A potential solution for this phenomenon was suggested in [1].

The 0-frequency adjustment can be modeled as follows

$$x'[i] = x[i] - h'$$

Where

$$h' = (\sum x[i]) / N$$

Essentially this adjustment makes note of the fact that the zero-frequency component is the simple average of the hydrophobicity values and therefore we just compute and subtract this component from the DFT.

Sample Set of TM proteins

Given the time constraints I only experimented with 19 TM proteins (from the PDBTM database) with 14 of them known to be dominated by alpha-helices and the remaining beta-dominated (see Table 1). Even though the list of proteins is small, the actual number of α and β strands in these sequences are quite significant. For example, the protein 1c17 (see Figure 1a) has 13 chains with each chain containing two distinct alpha-sequences, resulting in a total of 26 alpha-TM strands to experiment with (Table 1).

In the rest of this section I will refer to the sample-set dominated by alpha-structures as the *alpha-set* and the sample set dominated by beta structures as the *beta-set*. In addition, I will refer to the amino sub-sequences in my samples that were directly associated with

the alpha-helices as the *Alpha-strand*; on average these are about 20 residues long. Similarly the set of *Beta-strands* refers to the specific beta subsequences directly associated with the beta-sheets structures in the Beta-set; these on average are about 10 residues long. These statistics are also summarized in Table 1.

Analysis of Alpha-set

The frequency spectra for the alpha-set are displayed in Figure 2. While a visual inspection of these spectra does not reveal any obvious commonalities among them, some of the structural periodicities in the sequences can be inferred from them. These are described below for 2 specific cases, 1c17 and 2a9h.

1c17 - a1c12 subcomplex of F1F0 ATP Synthase (Figure 1a)

ATP synthases are membrane-bound ion transporters that are involved in the transport of protons across a membrane. F1F0 ATPases, a complex of 2 molecular motors, are ATP producers using a TM proton gradient to produce ATP. The 'a' subunit in this complex believed to mediate proton translocation across the membrane and the c_{12} oligomer which mechanically rotates around the 'a' subunit comprise a remarkable natural mechanical structure with complex moving parts involved in the ionic transportation of protons across a membrane. A spectral analysis of this protein (the 5th graph in Fig 2) reveals distinct peaks at frequencies 0.08, 0.19, and 0.38 that confirm some of the structural and sequence periodicity apparent in Fig 1a. For example the peak at around 0.08 can be explained by examining the 3-D structure in figure 1a. The sequence has 1140 amino acids of which 960 (80x12) constitute the 26 alpha-helices corresponding to a frequency of 0.08 (80/960). Other frequency peaks correspond similarly to specific periodicities of the hydrophobicity values. Not all of these periodicities appear to have direct analogs in the structural domain.

2a9h – K⁺ channel charybdotoxin complex (Figure 1b)

Ion channels across membranes play a critical role in signal transportation and electrolytic transfers and can therefore be an attractive area of research for treating diseases. Ionic channels are elegant gateways that rapidly conduct ions while at the same time being remarkably selective. The system can be generically described as comprising a pore, a selectivity filter, and a gate. As in the previous case, the spectral map (Fig 2) shows distinct peaks at frequencies 0.02, 0.1, 0.29, and 0.38. Its quite intriguing that 2a9h and 1c17 share spectral peaks at 0.1, 0.29, and 0.38. The common peak at 0.29 is especially remarkable and will be discussed further shortly.

A composite spectrum (construction methodology described in [17]) is shown in Figure 4b revealing two distinct peaks at 0.02 and 0.1 respectively. These frequencies correspond to residue periodicities of 50 and 10 respectively. However, a visual inspection of the 3-D structures, especially that of 1qhj whose spectrum most clearly shows these frequencies, does not reveal any obvious correlation between the frequency components and structural.

Analysis of Alpha Strands

As a second level analysis, the sub-sequences of the alpha-strands were extracted (a very manual and painstaking process!) and their frequency spectra computed (Figure 3). The spectra for these alpha-strands appear to be much cleaner and their composite waveform shows a strong frequency component at 0.29 (Figure 4a). This value corresponds to a hydrophobicity periodicity of around 3.5 and is consistent with the fact that alpha-helices typically have a structural periodicity of around 3.6 residues per turn.

Analysis of Beta Set and Beta Strands

The analysis of the beta-set was done in a similar way and is shown in Figure 5 with the composite spectra shown in Figure 7b. Just as with the Alpha TM proteins, I manually split each sequence to expose the beta strands and analyzed these sub-sequences separately. The frequency responses for the beta-strands are displayed in Figure 6 and the composite spectrum in Figure 7a. The individual waveforms in Figure 6 appear to be remarkably similar with a “characteristic” frequency around 0.46 corresponding to a residue periodicity of around 2.2. However, I hesitate to draw too strong a conclusion here because the beta strands are typically very small averaging around 10 in length (see Table 1) and this characteristic frequency pushes close to the Nyquist frequency of 0.5.

Summary Analysis

Comparing the composite spectra of the alpha and beta subsequences we see that the characteristics in one is largely absent in the other suggesting a way to identify such structures in a new previously unanalyzed TM sequence. It must be cautioned however that a naive application of these observations is ill-advised because some remnants of these periodicities are always embedded in generic sequences (compare Figure 2 and Figure 5)

As noted in the previous section, alternate experiments using different hydrophobicity scales, EIIP values, and even subdividing the amino acids into their constituent bases did not provide any additional insights into the frequency spectrum of these sequences.

Clearly, understanding the core function of TM proteins (such as 1c17 and 2a9h) and correlating them with their spectral patterns promises to be a rich source of future research, the scope of which unfortunately goes beyond the time constraints imposed on this term project.

Software and Platform

A short description of the software and tools used is described in the Appendix.

FUTURE WORK

From the above results, it appears that DFT analysis using hydrophobicity scales provide useful insights into TM structures that cannot be subsumed by probabilistic frameworks alone. Analysis of known 3-D structures of these proteins show high degrees of periodicity both at the macro level (in the high number of alpha and beta structures) as well as at the micro level (in analyzing the sub-sequences of these sub-structures as well).

Future work in this area can take several interesting directions. Given the small sample set used in this project, I envision the next steps would be to validate the results by extending the analysis to all TM structures in the TMPDB database of known structure.

A second project direction could be to do a deeper analysis of just TM protein (say the 1c17 or 2a9h) to better understand the relationship between the hydrophobicity periodicities and the structural periodicities. For example, why are the spectra in Figure 3 and Figure 5 so different? Is it mainly noise or is there hidden information (not well captured using a hydrophobicity scale) embedded that will help us better understand the TM structure?

Related to the above idea, better techniques need to be developed to accurately isolate the “hot” spots in the sequence responsible for a particular characteristic frequency. The methods described in [17] were attempted in this project with limited success. Alternately, a comprehensive prediction algorithm could bypass this issue altogether by splicing the sequence and analyzing the spectra for each subsequence. While this sidesteps the issue of finding hot spots it introduces the non-trivial problem of how to intelligently break up the sequence without losing valuable structural information.

As noted earlier, the quality of spectral analysis is only as good as the choice of the numerical sequence used to code the DFT. New ideas to encode the structural and chemical properties of residues can greatly improve the results we can obtain.

Finally, a spectral analysis tool will be incomplete unless it can be incorporated into a comprehensive structure prediction framework. The project work here had the benefit of working with TM proteins of known structure. Analyzing a completely new structure opens up many challenges. For instance, efficient heuristics must be developed to splice the sequence into sub-strands with a high probability of finding a secondary structure. DFT analysis such as that described here can then be used to further analyze these strands to confirm or deny the existence of alpha/beta structures.

PRIOR WORK

Structure Prediction Methods and Algorithms

While there have been numerous attempts at structure prediction from the amino acid sequence, accuracy has topped out at around 65-70% [4]. One of the earliest such methods, the Chou-Fasman method [9] looked at the frequency of occurrence of residues and ignored the interactions between pairs of amino acids.

More sophisticated techniques using neural-networks, nearest neighbor, and Hidden Markov Models (HMM) went a step further by analyzing the order of occurrence of amino acids in sequences. HMM-based approaches such as TMHMM [10] and HMMTOP[11] are based on a probabilistic framework consisting of a set of states corresponding to the general biological organization of the protein regions. PHDhtm[12] on the other hand leverages neural networks to predict TM regions from multiple-sequence alignments. However, these approaches essentially used a windowing technique to limit the “distance” of interactions to work around the NP-hard nature of the computing problem.

Hydropathy-based analyses [1][2][3] unlike statistical based methods rely on the physico-chemical properties of amino-acids and use hydrophobicity scales (developed from transfer-free energy and interior-exterior distribution of amino-acids) to predict transmembrane regions. The Kyte-Doolittle scale used in this project is an example of such a scale. Alternate scales were developed by computing the free energy for each of the 20 amino acids when they are transferred from water to oil by calculating the surface area of each amino acid side chain in an α -helix [22].

There have also been recent attempts to combine statistical methods (such as NN and HMM) with hydropathy analysis [13] to generate optimized hydrophobicity scales leading to improvements in the accuracy of the hydropathy analysis.

Signal-processing methods

DFT techniques for sequence analysis have grown in popularity in the last decade due to their elegance in capturing global patterns without expensive computational techniques. Some of the earliest work in this area involved defining indicator sequences for bases [16] and later demonstrating the period-3 property in base sequences using DFT techniques [14]. The sliding window technique used in these early analyses was shown to be a special case of digital filtering using a band-pass filter centered at 120° [15]. The method was further extended to use specially designed IIR Anti-Notch and multi-stage filters to isolate the $1/3$ frequency peak.

Spectral analysis for the study of the functional and structural relationships for a particular set of proteins was successfully performed using Resonant-recognition models [17]. That paper also showed how to predict the location of the biologically active sites allowing the possible design of synthetic peptides with those functions.

Numerical Coding sequences in Spectral Analysis

One of the most important questions in spectral analysis is the choice of the numerical coding sequence for the underlying amino-acid or nucleotide sequence. The main question to answer is what physical and chemical properties are accurately captured in the mapping.

Base sequence analysis has been historically performed using indicator sequences for the four individual bases followed by analyzing the amplitude of the composite sequences [16][18]. Improved results were obtained by analyzing the phase of the 1/3 frequency component using spectral rotation techniques to discriminate between coding and non-coding regions in the *S. Cerevisiae*, *S. pombe* and *Guillardia theta* [19]. A thorough DFT analysis of genomes using tri-color spectrograms was done [20] to clearly show various periodicities from single genes to whole chromosomes. This paper also served to highlight some of the potential risks of quickly jumping to conclusions related to periodicities and locations of exons.

An alternate popular approach has been the use of Electron Ion Interaction Potential (EIIP) [17] which describes the average energy states of all valence electrons in a given amino acid. Finally, the Hopp-Woods scale which denotes hydrophilic values was used along with a concept called a pseudo-amino acid composition to distinguish type I, type II, and chain-anchored membrane proteins [21].

Hydrophobicity Scales for Numerical Coding

Amino acids with non-polar side chains such as valine tend to be hydrophobic minimizing their contact with water by residing in the protein core. On the other hand, those with charged (lysine) and polar (serine) side chains reside on the surface where they can maximize their contact with water. These hydrophobic and hydrophilic tendencies play a crucial role in the formation of protein structures.

A large body of research has focused on the debate of how strong the hydrophobicity effect is on protein structure. Initially, it was shown that the distribution of hydrophobic residues is indistinguishable from that of a random distribution of soluble proteins [7]. However, later it was shown [1][2] using hydrophobic moments (as computed using hydrophobicity scales) that alpha helices show a strong periodicity of around 3.6 residues with hydrophobicity peaks at around 100° [2] (later revised to around 97.5° in [1]). Similarly beta sheets of known structure were demonstrated to have periodicity of around 160° or 2.3 residues per turn although the peaks for Beta sheets were much weaker than those of alpha helices. The debate was mostly resolved in 1995, when it was shown that hydropathy does in fact have a strong influence on a sequence's choice of alpha or beta structure [8].

Numerous scales have been developed over the years to measure the hydrophobicity/hydrophilicity of amino acids. Some scales [6] are derived from

examining proteins with known 3-dimensional structure and assigning values depending on an amino acid's propensity to be in the core, and others such as the Kyte&Doolittle scale [3], are derived from the physico-chemical properties (water solubility, effect on surface tension etc) of the amino acids. Due to the difference in their evaluation schemes, the scales vary significantly in their scoring of the amino acids. A rigorous analysis of 38 of such scales was done in [1] using the concept of an amphipathic index to evaluate the predictive quality of the various scales.

CONCLUSIONS

In this project, I have explored the question of whether spectral data can guide the structural and functional analyses of Trans-membrane (TM) proteins. I have produced some initial results using DFT techniques and hydrophobicity scales that demonstrate that TM regions comprising alpha-helices have characteristic frequency components very distinct from those seen in beta-dominated TM regions. In addition, I have confirmed that the characteristic frequencies of the alpha and beta strands are around .29 and .46 respectively. These frequencies appear to correlate quite well with the structural periodicities of 3.6 and 2.3 residues per turn observed in these structures.

An attempt has also been made to better understand the relationship between the frequency spectrum and the structural and functional properties of two specific proteins – 1c17 and 2a9h. For example, the existence of the 26 alpha-TM regions in 1c17 manifests itself as a distinct peak in the frequency spectrum.

Finally, the project results suggested new avenues for research. These include performing an exhaustive analysis of all known TM protein structures; detailed analysis of the spectrum of a single TM protein to better understand which frequency components are due to noise and which embed real insights; New research in the choice of numerical scales for the DFT mapping; and finally incorporating DFT techniques along with probabilistic and hydropathy-based analysis into a comprehensive structure prediction framework.

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APPENDIX

(contains figures, tables, and notes on software and platform)

Table 1: Sample Set of TM Proteins

Type	PDB_ID	Number of TM strands	Avg amino acids per α and β strand
Alpha			
	1c17	26	19
	1a11	1	18
	1afo	2	22
	1ar1	14	19
	1kju	10	18
	1qhj	21	21
	1rc2	6	18
	2a65	12	22
	2a9h	8	24
	2ahy	8	19
	2akh	30	12
	2rcr	11	17
	1bts	1	15
	1dxz	1	22
Beta			
	1a0t	54	9
	1bxw	8	10
	1by3	22	6
	1ek9	12	16
	2erv	16	9

Table 2: K&D Hydrophobicity Scale Used for coding the DFT

Amino Acid Codes	Numerical Code
Alanine (A)	1.8
Cysteine (C)	2.5
Aspartic Acid (D)	-3.5
Glutamic Acid (E)	-3.5
Phenylalanine (F)	2.8
Glycine (G)	-0.4
Histidine (H)	-3.2
Isoleucine (I)	4.5
Lysine (K)	-3.9
Leucine (L)	3.8
Methionine (M)	1.9
Asparagine (N)	-3.5
Proline (P)	-1.6
Glutamine (Q)	-3.5
Arginine (R)	-4.5
Serine (S)	-0.8
Threonine (T)	-0.7
Valine (V)	4.2
Tryptophan (W)	-0.9
undetermined (X)	0
Tyrosine (Y)	-1.3
either glutamic acid or glutamine (Z)	-3.5

Figure 1a: Structure of 1c17 courtesy of PDBTM (<http://pdbtm.enzim.hu>) and JMol

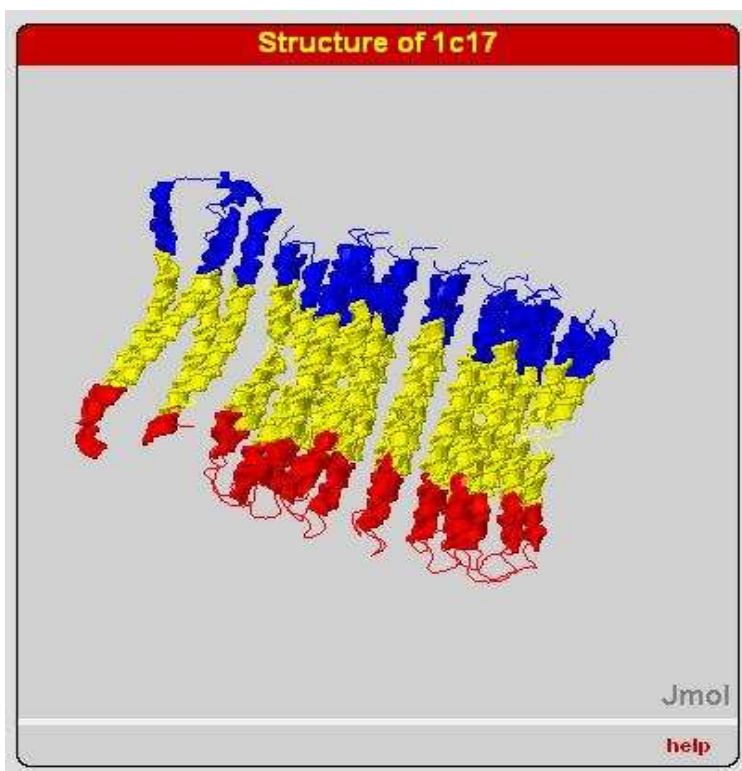


Figure 1b: Structure of 2a9h courtesy of PDBTM (<http://pdbtm.enzim.hu>) and JMol

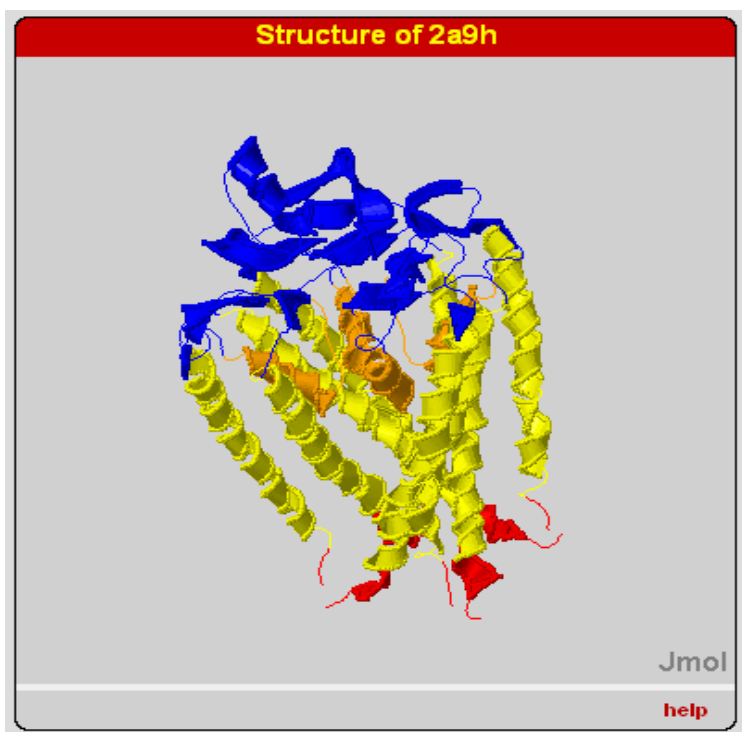


Figure 2: Spectra of the alpha TM proteins (only first 9 displayed)

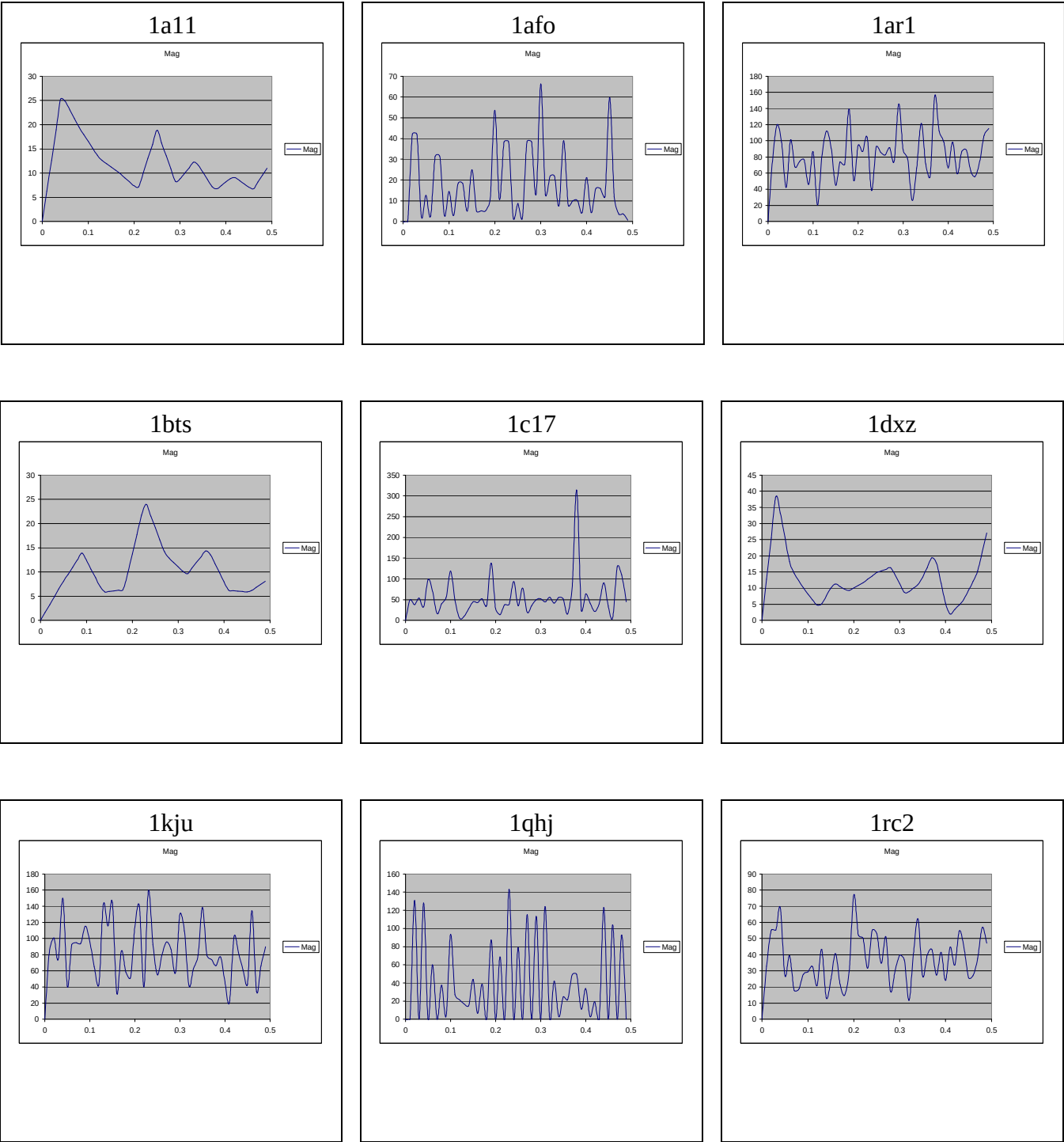
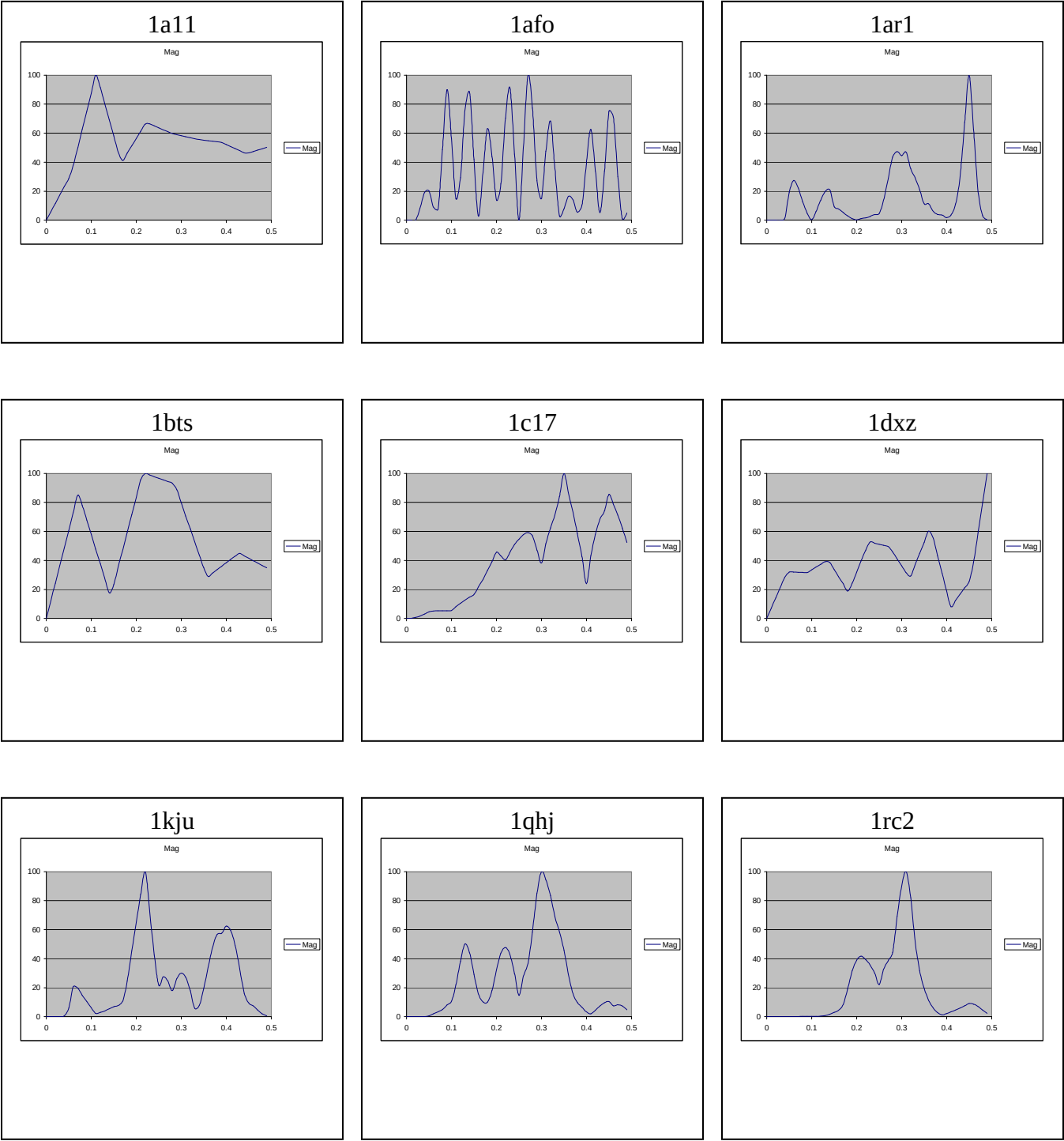
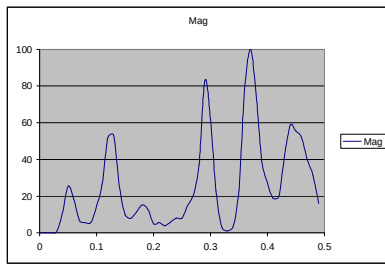


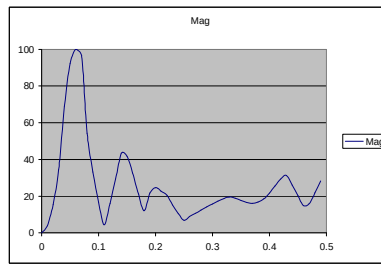
Figure 3: Spectra of just the Alpha strands of alpha TM proteins



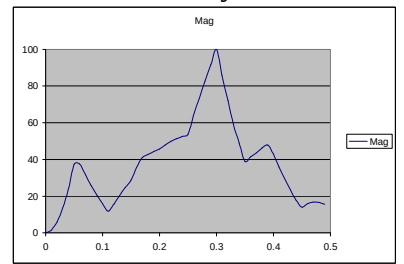
2a65



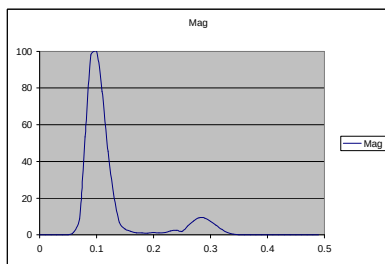
2a9h



2ahy



2akh



2rcr

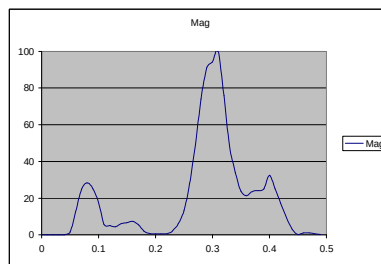


Figure 4: Composite DFT spectra for the alpha TM proteins (a) Shows the DFT after isolating the alpha strands (around 20 residues in length) from the TM proteins. There's a sharp peak at frequency 0.29. (b) DFT for the entire sequence. Sharp peaks can be observed at 0.02 and 0.1

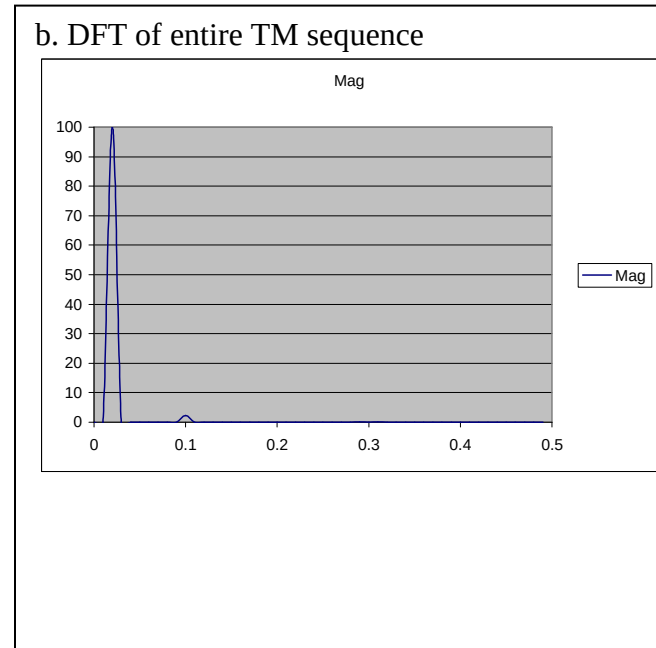
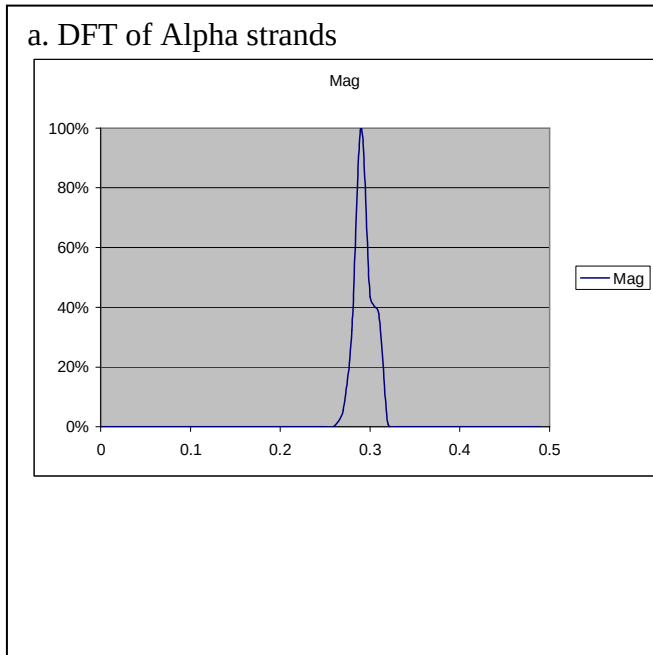


Figure 5: Spectra of the Beta TM proteins.

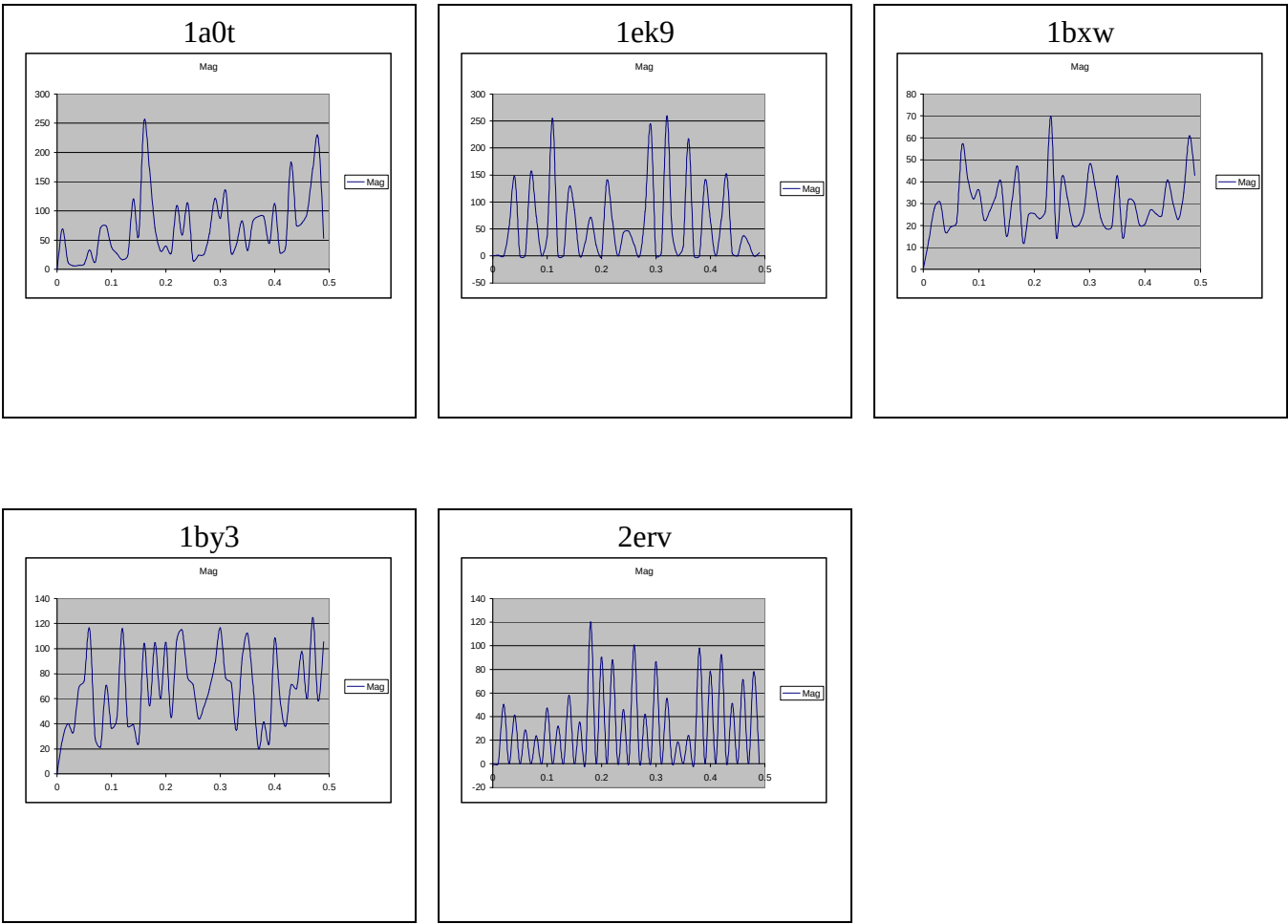


Figure 6: Spectra of just the Beta strands of the Beta TM proteins.

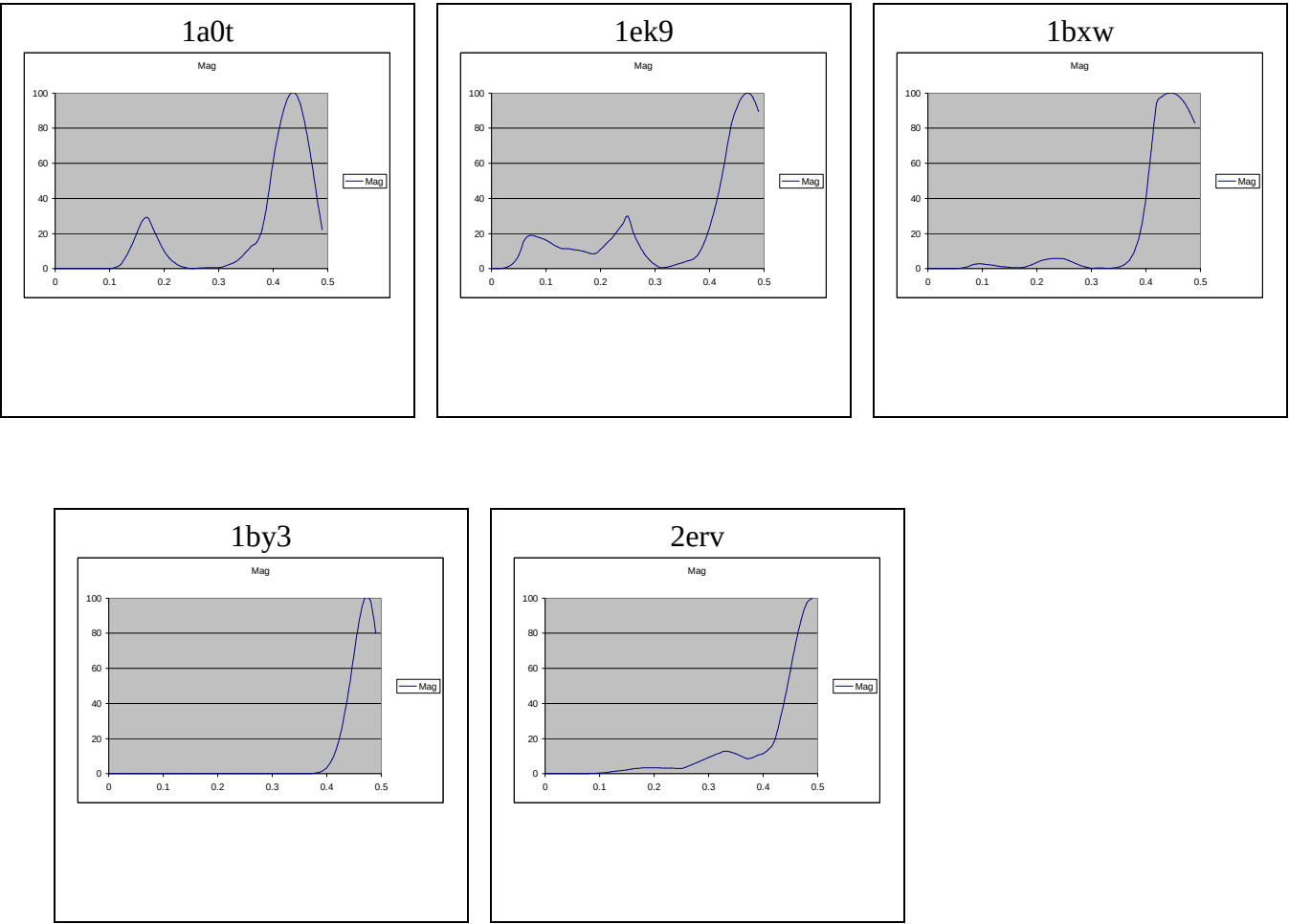


Figure 7: Composite DFT spectra for the beta TM proteins (a) Shows the DFT after isolating the beta strands (around 10 residues in length) from the TM proteins. There's a sharp peak at frequency 0.46. (b) DFT for the entire sequence. Many peaks can be observed

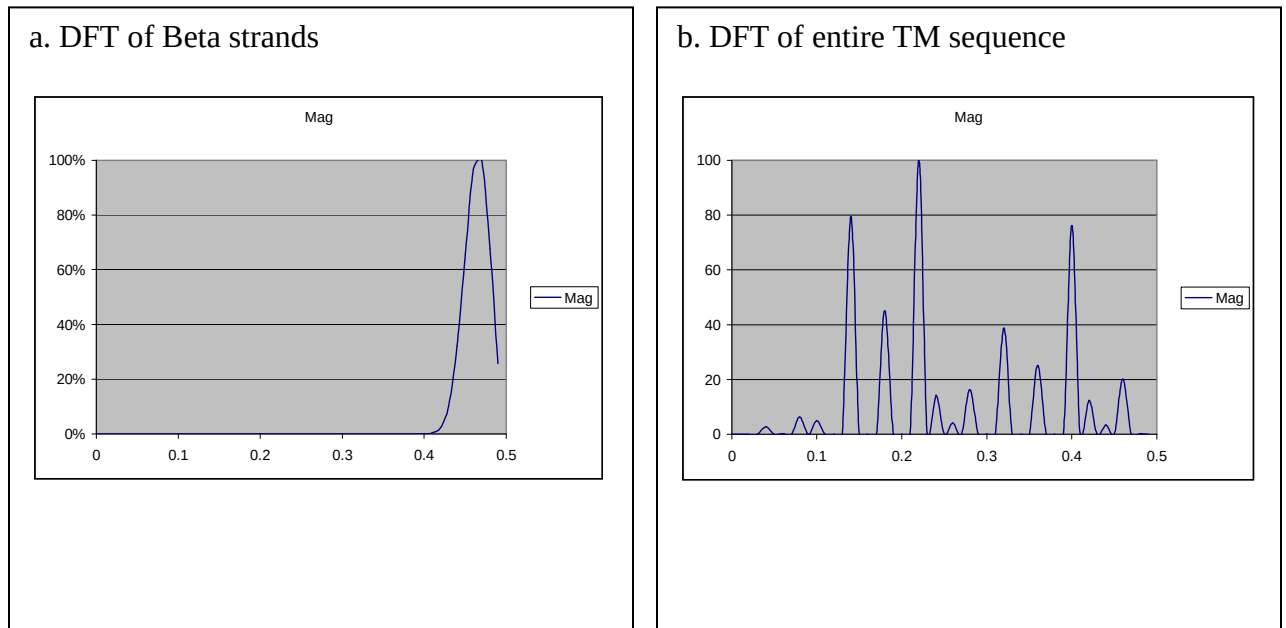
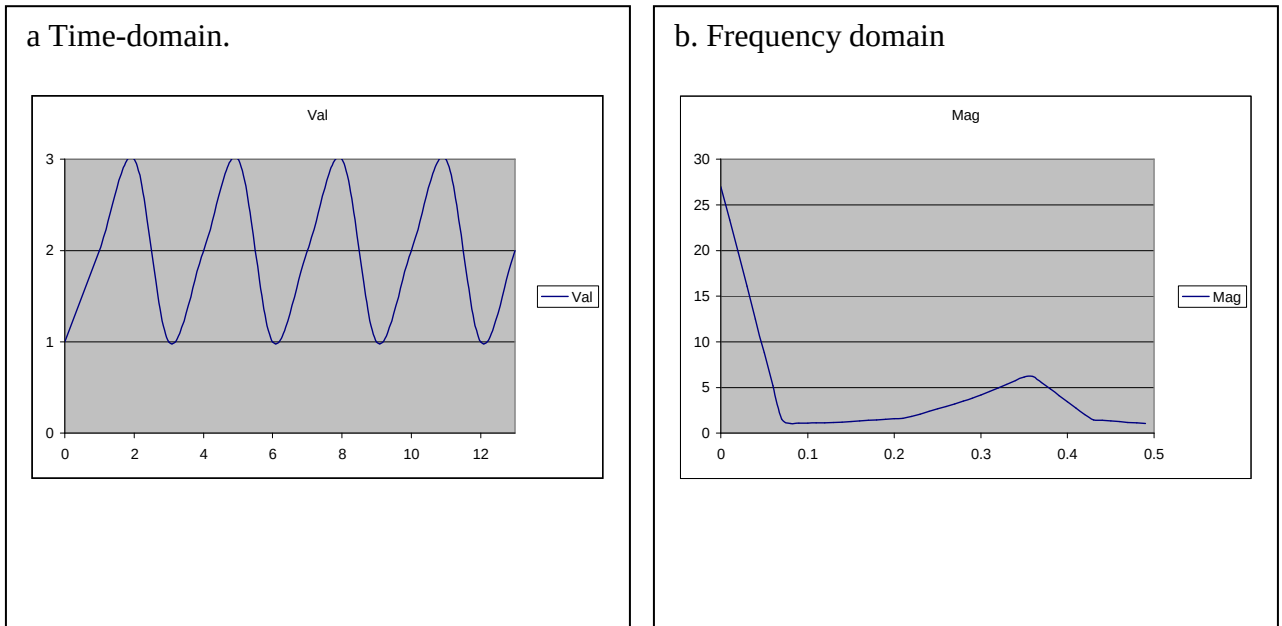


Figure 8: Demonstration of zero-frequency component in DFT. Time-domain signal is a saw-tooth periodic signal (period 3). The frequency spectrum shows a clear peak at 0.33 which is strongly dominated by the frequency component at 0



Software and Platform

This project was implemented on a Windows platform using Activestate Perl. The sequences were downloaded from the Protein Data Bank (<http://www.rcsb.org/pdb/home>). Visual inspection of the various TM structures was performed with the aid of the tools available at the PDBTM website (<http://pdbtm.enzim.hu/>).

The DFT source code implemented as part of this project is completely standalone, using no third-party modules and is portable to other platforms. The code is available upon request.